
ConceptFormer-2: Token-Efficient Grounding of Frozen Instruction-Tuned Language Models in Knowledge Graphs

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Abstract

We present ConceptFormer-2, which grounds a *frozen*, instruction-tuned language model in a knowledge graph by compressing an entity’s 1-hop neighborhood into k continuous *concept tokens* spliced into the model’s input embeddings — replacing ~ 100 tokens of retrieved text with $k \approx 8$. The encoder, a small Perceiver-style resampler over the frozen model’s own label embeddings, is inductive (unseen entities require one forward pass) and trains *label-free* by full-vocabulary KL self-distillation against the same frozen model reading the facts as text. On the full 14,266-question PopQA benchmark, $k=8$ concept tokens lift a frozen Qwen3-0.6B from 0.103 to 0.477 exact-match on unseen entities — $2.6\text{--}3.2\times$ better than question-aware retrieval or LLM-written summaries at the same token budget — while an untrained-injection control stays near floor, showing the learned compression carries the effect. Accuracy rises monotonically in k through $k=32$ on matched curves at two corpus scales, exposing a three-way trade between token budget, training data, and model size: unseen-entity accuracy grows $2.3\times$ when training entities grow $10\times$, and additional tokens partially substitute for additional data. Counterfactual edge-swap probes show the model genuinely *reads* the injected graph, increasingly so with k , while off-topic behavior is preserved (median next-token KL ≤ 0.08 nats). ConceptFormer-2 turns a frozen conversational model into an entity-grounded QA system at roughly one-tenth the context cost of retrieval, with all corpora, checkpoints, and per-item evaluation outputs released.

1 Introduction

Instruction-tuned language models answer questions about popular entities from their weights and fail on the long tail — a deficit that persists even in frontier models, which encode far more long-tail facts than they can recall [20, 21]. The standard remedy, retrieval-augmented generation [22], restores factuality but pays for it on every query: a single entity’s verbalized Wikidata neighborhood occupies a median of ~ 100 context tokens in our corpora, competing with the user’s actual task for context budget, latency, and key-value cache.

ConceptFormer [1] proposed a third channel: compress the knowledge-graph neighborhood itself into a few *continuous* tokens that a frozen LM consumes natively. The original system was validated

*Draft author list and contributions preliminary. All numbers trace to per-item evaluation dumps and W&B run ids in the project repository (joelbarmettlerUZH/ConceptFormer, branch conceptformer-v2).

on GPT-2 (0.1B) with rank-based metrics on next-token factual recall and received the best paper award of its hosting workshop at The Web Conference 2026. It left the harder questions open: does latent injection survive *generative, exact-match QA with a modern conversational model*? Does the model actually *read* the injected structure, or merely absorb a text-like signal? And — decisive for whether the channel matters at scale — what governs its value: the token budget k , the amount of knowledge the encoder is trained on, or the size of the frozen model?

ConceptFormer-2 answers these questions with a redesigned system and a measurement-first methodology. Three design changes carry most of the weight: (i) an *inductive* featurizer that represents each graph edge with the frozen model’s own label embeddings — replacing v1’s per-entity lookup table, so unseen entities need labels, not pretrained vectors; (ii) a *label-free* training objective — full-vocabulary KL distillation of the frozen model reading k concept tokens against the *same* frozen model reading the facts as text — requiring no annotated QA anywhere; and (iii) a zero-initialized output projection that makes the student bit-identical to the frozen model at step 0, preserving capability by construction.

Contributions. (1) A modernized latent knowledge-injection system for frozen instruction-tuned LLMs, evaluated with generative exact match on the full PopQA benchmark, with strong text baselines (question-aware retrieval, LLM-written budgeted summaries) and an untrained-injection control (§5.1). (2) Matched token-budget curves at two corpus scales showing monotone gains through $k=32$ and a three-way trade structure: data scales injection up ($2.3\times$ for $10\times$ entities), tokens partially substitute for data, and we measure the frozen-model-size axis across the Qwen3 family (§5.2). (3) Causal evidence that the model reads the injected graph — counterfactual edge-swap following that scales from $\sim 1\%$ ($k=1$) to $\sim 34\%$ ($k=32$) — alongside capability preservation and prompt robustness (§5.3). (4) An open release: corpora with per-teacher distillation targets, all checkpoints, the evaluation harness, and per-item outputs enabling third-party paired re-analysis.

2 Related work

Lineage. The problem framing descends from language-models-as-knowledge-bases probing [23, 24] and its long-tail refinement [21, 20], with RAG [22] as the deployed remedy. Knowledge-enhanced PLMs inject KG signal by modifying the model — KnowBERT [25], K-BERT [26], K-Adapter [27] — which our frozen-model constraint rules out. ConceptFormer v1 [1] injected *pre-trained* KG node embeddings (TransE-family vectors at PyTorch-BigGraph scale [28, 29]) through a per-entity lookup table; the present system replaces this with features built from the frozen model’s own label embeddings, which is what makes the encoder inductive. Prepending *learned continuous tokens* to a frozen model descends from prefix-tuning and soft prompts [30–32], textual inversion [33], and most directly Frozen [34] and Flamingo [36]. Our substrate is Wikidata [37]; graph verbalization baselines follow the graph-to-text encoding line [38].

Soft-token context compression. Gist tokens [4] and AutoCompressor [5] adapt the LM itself; ICAE [3] and 500xCompressor [6] train LoRA encoders against frozen decoders with reconstruction objectives; CompLLM [8] distills per-segment embeddings via activation matching; ARC-Encoder [9] is the strongest recent frozen-decoder text compressor. PISCO [7] shares our sequence-distillation family but reports that frozen decoders underperform (their Appendix G) — a claim our frozen-decoder results contradict at entity granularity. Closest is xRAG [2]: frozen decoder, input-embedding injection, same-model KL distillation — but xRAG *translates* a single pre-existing retriever vector rather than learning compression, is retriever-bound at inference, and evaluates neither unseen-entity generalization nor faithfulness. Cartridges [10] distills a KV cache *per corpus* by gradient descent: no amortization, no unseen items. None of these compress *structured* input, and none runs causal probes on the compressed representation (post-hoc overflow probing [11] being the nearest attempt).

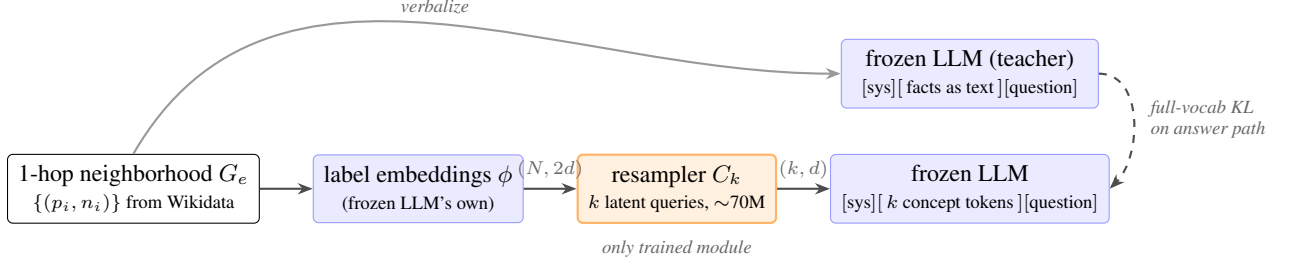


Figure 1: ConceptFormer-2. Edge features are the frozen model’s own mean-pooled label embeddings (inductive: any labeled entity works); a Perceiver-style resampler compresses the edge set into k concept tokens spliced into the frozen model’s input. Training minimizes the full-vocabulary KL between the same frozen model reading facts as text (teacher) and reading k tokens (student), along the teacher’s greedy answer path — no QA labels, no LLM updates.

Knowledge-graph injection into LLMs. GNP [14], GraphToken [15], G-Retriever [16], and LLaGA [17] encode graphs into soft prompts *per question* with task cross-entropy on labeled QA; none precomputes reusable per-entity tokens. KBLaM [12] precomputes one key–value pair per *triple*, injected into every attention layer through modified attention, evaluated on synthetic KBs. Knowledge Prompts [13] is the closest precompute precedent — per-entity soft prompts for 1.1M Wikidata entities — but as a lookup table of free parameters, making unseen entities impossible by construction. Recent analyses find graph-token LLMs “do not fully understand graph tokens” [18] and that graph tokens can decouple from graph semantics [19]; our counterfactual evaluation (§5.3) is designed to measure exactly this. No prior work combines an amortized *inductive* per-entity encoder, label-free same-model distillation, unseen-entity and causal evaluation, and corpus scale.

3 Method

Setup and featurizer. Let M be a frozen chat LLM with input-embedding dimension d . For an entity e with 1-hop neighborhood $G_e = \{(p_i, n_i)\}_{i=1}^N$ (property and neighbor labels, PageRank-ordered), each edge becomes $x_i = [\phi(p_i); \phi(n_i)] \in \mathbb{R}^{2d}$, where ϕ mean-pools M ’s input embeddings over a label’s tokens. Features are inductive and live in a space M already understands; no per-entity parameters exist anywhere.

Encoder. A latent-query resampler [35, 36]: k learned queries cross-attend over the masked, permutation-invariant edge set, self-attend, and pass through a feed-forward block, for L layers ($d_{\text{model}}=1024$, $L=4$, $\sim 70M$ parameters — an encoder capacity ablation appears in Table 2); an output projection maps to $\mathbb{R}^{k \times d}$. The projection is *zero-initialized*, so at step 0 the student is bit-identical to the frozen model; we found this variant more stable than a multiplicative gate.

Objective. The teacher is M reading [system][verbalized G_e][question]; its greedy answer path $y_{1..m}$ is stored offline. The student is M reading [system][$C_k(G_e)$][question], with the concept block occupying a contiguous unit-step RoPE span. The loss is the token-level full-vocabulary forward KL along the stored path,

$$\mathcal{L} = \frac{\tau^2}{m} \sum_{t=1}^m \text{KL}(p_M(\cdot \mid \text{facts}, q, y_{<t}) \parallel p_M(\cdot \mid C_k(G_e), q, y_{<t})), \quad \tau = 1. \quad (1)$$

Only the ~ 20 path positions pass through the LM head, keeping training memory-light; one converged run fits a single 24 GB consumer GPU. Corpus questions (generated by a separate LLM from the graphs; below) exist solely to give the teacher something to answer — no QA labels enter the loss.

Data. From the Wikidata PageRank ranking we sample entity pools (PopQA subjects excluded), snapshot complete 1-hop neighborhoods, generate questions with an open-weights Gemma model, *tier* each question by whether the graph demonstrably helps the backbone (base fails, facts-in-context succeeds; a 30% sample of base-known questions is retained), and store teacher paths. Corpora: 10k entities (92,177 rows under the Qwen3-0.6B teacher) and 100k entities (925,178 rows). Tiering and teacher paths are teacher-relative, so each backbone trains on its own corpus variant of near-identical size — comparisons across backbones are of “each model with its own curriculum”.

Training. AdamW, lr 10^{-4} held constant, weight decay 0.01, effective batch 32 via gradient accumulation, 72k optimizer steps on the 10k corpus and 100k steps (≈ 4 epochs) on the 100k corpus, best-checkpoint selection on a held-out sample. Every reported configuration uses ≥ 3 random seeds.

4 Evaluation protocol

All results are bracketed by two conditions of the same frozen model: **base** (no knowledge; the floor) and **RAG** (facts as text with the questioned edge guaranteed present; the ceiling, 0.96 on full PopQA). The metric is greedy exact match with word-boundary alias matching; we additionally report the official (more lenient) PopQA metric, which tracks ours within ~ 0.5 pt. Three accuracy axes: **held-out** (unseen questions about trained entities), **held-in** (trained questions; an overfitting gauge), and **PopQA** (unseen *entities* — the axis that requires inductive generalization and is the paper’s headline).

Four protocol properties matter for the statistics we report. (1) PopQA is evaluated on the *full* benchmark ($n=14,266$ questions with snapshot coverage), bringing binomial noise to ~ 0.4 pt. (2) Held-out splits are *fact-level*: paraphrases of the same (entity, fact) never straddle train/test, and held-out sets are additionally filtered so no tested fact was supervised. (3) Evaluation subsets are frozen and shared across all runs, configurations, and seeds, so any two systems can be compared item-by-item; we report Wilson intervals and exact McNemar tests on shared items. (4) Definitive numbers come from re-scoring selected checkpoints on large, disjointly-sampled sets, never from the trajectory samples used for checkpoint selection. Per-item outputs for every evaluation are released.

5 Results

5.1 Token efficiency against strong text baselines

Figure 2 compares concept tokens against three text baselines at matched knowledge-token budgets: query-independent top-PageRank truncation, *question-aware* retrieval (facts ranked by embedding similarity to the question — an advantage our query-independent tokens do not have), and LLM-written budgeted summaries. At an 8-token budget on unseen entities, concept tokens reach 0.477 where the best text baseline reaches 0.147; at 16 tokens, 0.518 vs. 0.243. Text requires 64–128 tokens for parity, i.e., matched accuracy costs text $\sim 5\text{--}6\times$ more tokens. Two controls anchor the comparison. First, filling the same k slots with *untrained* mean-pooled embeddings of the top- k edges scores 0.130 on PopQA (base 0.103), identically for $k=8$ and 16: the learned compression carries $\sim 93\%$ of the injected-knowledge effect. Second, the teacher/RAG ceiling (0.960) confirms ample headroom remains.

5.2 The three-way trade: tokens, data, and model size

Token axis. Table 1 and Figure 3: accuracy is monotone in k through $k=32$ on *both* corpora, with every adjacent contrast item-paired significant at 100k. Returns per token diminish — $k=8$ buys $\sim 89\%$ of $k=32$ ’s PopQA at a quarter of the tokens — so the efficiency-optimal budget is a deployment choice, not a capacity cliff.

Data axis. Scaling training entities $10\times$ (identical eval sets, matched placement, $k=8$) lifts full-PopQA accuracy from 0.210 ± 0.012 to 0.477 ± 0.002 — a $2.3\times$ gain in unseen-entity generalization

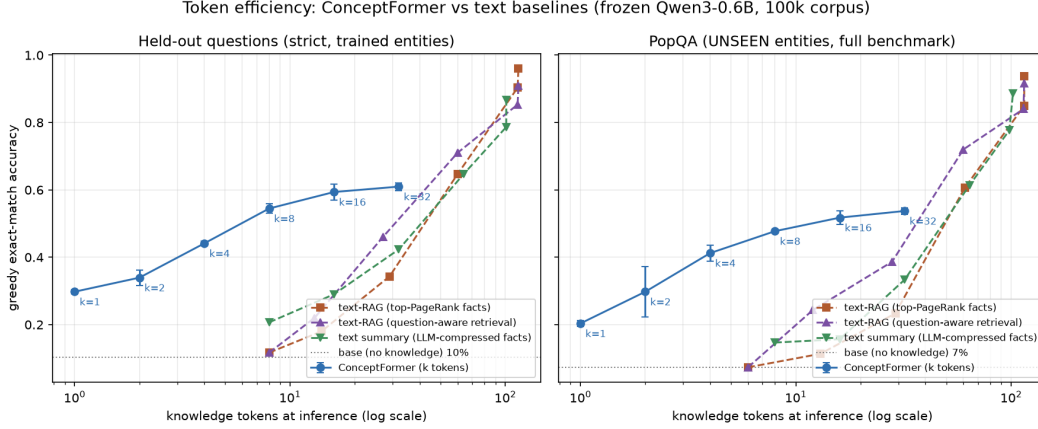


Figure 2: Accuracy vs. knowledge tokens spent at inference (log scale), frozen Qwen3-0.6B, 100k-entity corpus, 3 seeds. Concept tokens dominate all three text baselines below ~ 32 tokens on both axes; text catches up only at 64–128 tokens (most of a verbalized neighborhood). Left: strict held-out; right: full PopQA.

Table 1: Corrected k -family on the 100k-entity corpus (frozen Qwen3-0.6B, 3 seeds, mean $\pm$ std). Strict held-out $n=2000$; full PopQA $n=14,266$; official PopQA metric for literature comparability. Brackets: base 0.103 (official 0.107), RAG 0.960. All adjacent k -contrasts are significant under item-paired McNemar (largest $p \approx 10^{-6}$).

k	1	2	4	8	16	32
held-out	.298 $\pm$.006	.340 $\pm$.023	.441 $\pm$.007	.545 $\pm$.014	.594 $\pm$.024	.610 $\pm$.011
PopQA	.203 $\pm$.008	.298 $\pm$.075	.412 $\pm$.023	.477 $\pm$.002	.518 $\pm$.020	.537 $\pm$.010
PopQA (official)	.207 $\pm$.009	.303 $\pm$.077	.419 $\pm$.024	.483 $\pm$.002	.523 $\pm$.021	.543 $\pm$.010

that had not saturated at the end of training. Held-out rises too ($0.416 \rightarrow 0.545$) while held-*in* falls ($\sim 0.82 \rightarrow \sim 0.64$): with more entities the encoder memorizes individual training entities less and learns the edge $\rightarrow$ concept mapping more — the signature of graph-learning rather than corpus memorization.

Token–data substitution. The matched curves expose an interaction: the 10k $\rightarrow$ 100k PopQA ratio falls from $2.05\times$ ($k=8$) to $1.68\times$ ($k=16$) to $1.48\times$ ($k=32$). A larger token budget extracts more value from a small training corpus, partially substituting for data — consistent with a rate–distortion view in which both k and corpus size bound how much neighborhood structure the encoder can faithfully transmit.

Model axis. We repeat the data transect across the Qwen3 family (0.6B, 1.7B, 4B), each backbone with its own teacher-relative corpus and brackets (Figure 4). [Pending cells: 1.7B $\times$ {10k, 100k} and 4B $\times$ {10k, 100k} at $k=8$ (and $k=16$ at 1.7B), 3 seeds each — training in progress; this subsection’s claims will be finalized from those cells.] The question the grid answers: does a larger frozen model — which knows more of PopQA parametrically — gain less from injection at fixed data, and if so, does more data close the gap?

5.3 Does it read the graph? Causal probes

A compression system can score well on QA while treating the graph as an opaque text-like signal. Following the deficiency diagnosed across graph-token systems [18, 19], we evaluate causally, by intervening on the *input graph* of a trained model and re-encoding: (a) *edge ablation* — removing the questioned edge collapses accuracy on that question (to 28–55% depending on k) while removing an unrelated edge leaves it intact (90–96%): edges are encoded separately; (b) *counterfactual swap*

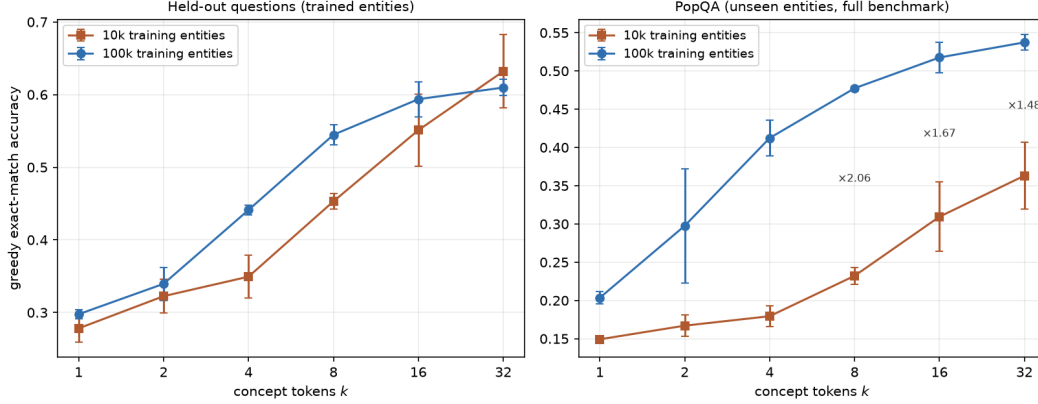


Figure 3: Matched token-budget curves at two corpus scales (3 seeds, identical protocol). Accuracy rises monotonically in k through $k=32$ at both scales. Annotations on the right panel give the 10k $\rightarrow$ 100k PopQA ratio per k : the data-scaling gain shrinks as k grows — additional tokens partially substitute for additional training data.

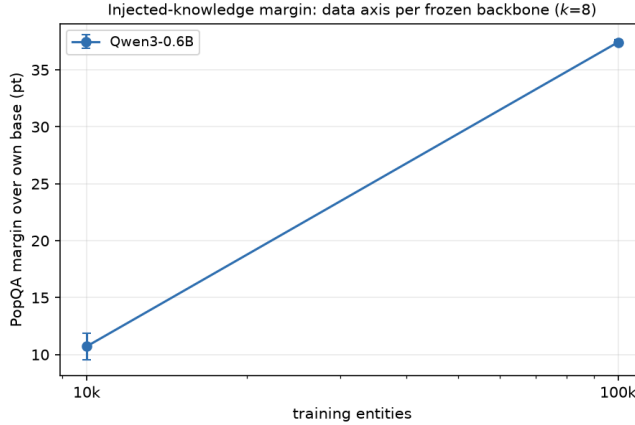


Figure 4: The data $\times$ model grid ($k=8$): PopQA margin over each frozen backbone’s own base accuracy. [1.7B and 4B cells land as training completes; figure regenerates from the same script.]

— rewiring the questioned edge to a plausible false neighbor flips the model to the *false* answer with probability rising monotonically from $\sim 1\%$ at $k=1$ to $\sim 34\%$ at $k=32$ (Figure 5, left; 3 seeds). Since the frozen base solves only 10% of these probes, parametric memory cannot explain the behavior: the concept tokens are causally load-bearing, and faithfulness is *capacity-limited rather than absent*. Meanwhile injection does not disturb the model elsewhere: on off-topic control tasks the median next-token KL between the model with and without concepts stays at 0.05–0.08 nats, flat in k (Figure 5, right) — by construction, since the student starts bit-identical to the frozen model. Accuracy is also robust to prompt wording: across seven system prompts (five seen styles, one held-out), PopQA accuracy of the $k=8$ model varies with a standard deviation of ≤ 1.1 pt across 3 seeds.

5.4 Ablations

Table 2: encoder capacity peaks near 70M parameters with the 231M encoder clearly worst at this budget; placing concepts adjacent to (or in place of) the entity mention does not beat a simple prefix; multi-prompt augmentation yields a small but, under pooled paired testing, significant gain with real seed heterogeneity; per-step re-sampling of teacher distractors is a null at our context budget.

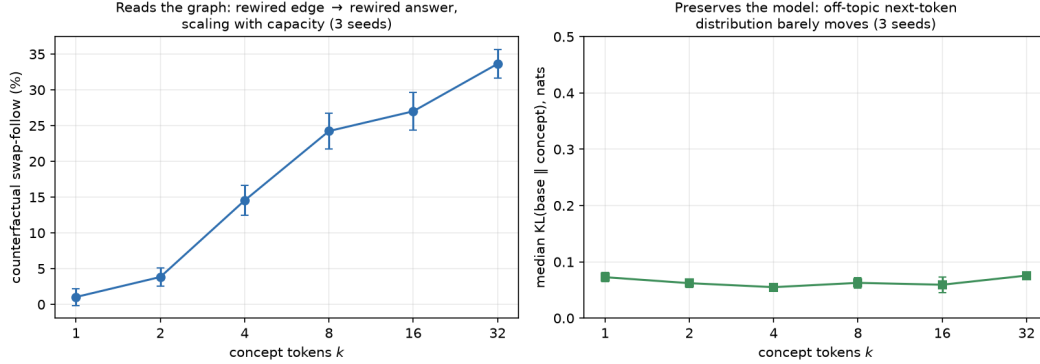


Figure 5: Causal probes across k (Qwen3-0.6B, 100k corpus, 400 single-fact probes per checkpoint, 3 seeds). Left: after rewiring the questioned edge to a type-plausible false neighbor, the model increasingly follows the graph to the *false* answer — the graph-reading signature, absent any parametric shortcut (the frozen base knows only 10% of these probes). Right: on control tasks that mention the entity without asking about its facts, the next-token distribution barely moves, flat in k .

Table 2: Design ablations (10k corpus, $k=8$, 3 seeds each, strict held-out / full PopQA). Augmentation: pooled item-paired McNemar over 3 seeds; effect positive on average with seed-level heterogeneity (one seed reverses on PopQA).

ablation	variants	held-out	note
encoder capacity	10M / 21M / 70M / 231M	.423 / .441 / .453 / .376	peak at ~ 70 M; largest is worst
placement	prefix / before / after / replace	.453 / .423 / .434 / .380	entity-adjacency does not help
prompt augmentation	off / on	.453 / .472	PopQA +2.0 pt; pooled $p=3 \times 10^{-36}$
neighbor subsampling	off / on	$\Delta \approx 1$ pt	null; cached teacher is $2\times$ faster

6 Discussion

What governs the channel’s value. The results organize around a three-way trade. Token budget buys accuracy monotonically but with diminishing returns; training data buys unseen-entity generalization multiplicatively and had not saturated; and the two interact — tokens substitute for data. The model-size axis (Figure 4) determines whether these gains persist as the frozen model itself knows more; the two live interpretations — a *ceiling effect* (bigger bases leave less headroom on the benchmark) versus a *steerability effect* (bigger frozen models need more training signal per unit of behavioral change from k soft tokens) — make opposite predictions about whether more data closes the gap, which the grid’s 100k cells test directly. [Finalize once 1.7B/4B cells land.]

Limitations. Single knowledge graph (Wikidata) and language (English); entity-valued relations only (no literals); evaluation centered on PopQA and same-generator held-out questions; faithfulness probes cover single-fact questions; corpora are teacher-relative by design, so cross-backbone comparisons involve each model’s own curriculum; augmentation’s effect is heterogeneous across seeds. Knowledge is injected for a single identified entity mention per prompt; multi-entity composition is untested.

Outlook. The encoder is modality-agnostic by construction — its inputs are embeddings, not text — and extending concept tokens to multimodal knowledge (e.g., entity images) for natively multimodal frozen models is the subject of ongoing work.

7 Conclusion

ConceptFormer-2 shows that a frozen, instruction-tuned language model can be grounded in a knowledge graph through $k \approx 8$ learned continuous tokens: $4.6\times$ over the frozen base on unseen entities

at one-tenth of retrieval’s token cost, with causal evidence that the graph is genuinely read and the model otherwise undisturbed. The value of the channel has measurable structure — monotone in tokens, multiplicative in data, interacting between the two — and the released corpora, checkpoints, and per-item evaluations make every number in this paper re-derivable and extensible.

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